

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Metallurgical & Materials Engineering

2. Subject Name: Machine learning methods for materials engineering

L-T-P: 3-0-0

Credits: 3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

(1) Overview of machine learning and materials informatics: Fundamental concepts of ML; Regression vs. classification; Supervised and unsupervised learning; Parametric vs. non-parametric models; Bias-variance trade-off, etc.; applications of informatics methods in materials engineering.

(2) Mathematical preliminaries: Review of linear algebra and statistics relevant to machine learning.

(3) Mathematical and numerical optimization: Global and local extrema; constrained and unconstrained optimization; Numerical optimization with first and second order methods.

(4) Feature extraction and dimensionality reduction: Use of correlation maps; PCA and LDA; Isomap and Locally linear embedding; A relevant case study like classification of metallic glasses using PCA.

(5) Linear regression and classification models: Linear regression; Logistic regression; Support vector machines and its variations; Case studies of band-gap engineering for solar power materials through regression analysis and SVM-based phase prediction of multi-principal element alloys.

(6) General learning of discriminative models: Loss functions in machine learning; Ridge regression and LASSO; Matrix factorization; case study of predicting glass forming ability through LASSO.

(7) Neural networks: Basic formulation and mathematical justification of ANNs; Basic building blocks and structures; Fully connected deep neural networks; convolutional and recurrent NNs; learning algorithms; case studies of tensile strength prediction through ANNs and recognizing ceramic microstructure using CNNs.

(8) Ensemble learning: Basic formulation; Decision trees; Random forests; Boosting algorithms; Case study of predicting yield strength of high entropy alloys using random forest regressor.

Note: The case studies mentioned above are for demonstration. The instructor(s) can replace them with other examples in view of convenience and/or availability of more

updated examples. It is recommended that around 30% of all the available contact hours must be utilized for the case studies. **Example cases should preferably involve a variety of systems like alloys, semiconductors, ceramics, polymers, etc., to showcase the wide applicability of the methods.**

Reference Books:

- 1) H. Jiang, *Machine learning fundamentals: A concise introduction*, Cambridge University Press.
- 2) K. Rajan (ed.), *Informatics for materials science and engineering*, Elsevier.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures/tutorials
1.	Overview of machine learning and materials informatics	2
2.	Mathematical preliminaries	4
3.	Mathematical and numerical optimization	4
4.	Feature extraction and dimensionality reduction	4
5.	Linear regression and classification models	6
6.	General learning of discriminative models	5
7.	Neural networks	6
8.	Ensemble learning	5
Total number of hours		36